MANIPAL INSTITUTE OF TECHNOLOGY
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UrjaGrid Monitor

Solar Microgrid Monitoring and Independent Load Control

Final Mini Project Report

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Abstract

UrjaGrid Monitor is an educational solar microgrid prototype that combines individual load measurements, remote switching and battery-aware load prioritization. A solar panel and charge controller support a 12 V, 7.2 Ah VRLA battery supplying a DC bulb and fan. An ESP32 reads two independently addressed INA219 modules and a DS18B20 temperature sensor, then exchanges telemetry and acknowledged commands with a FastAPI backend hosted on Vercel. Supabase PostgreSQL stores measurements and activity records for historical graphs and energy estimates. Automatic protection disconnects the lower-priority fan below 50 percent voltage-estimated state of charge, permits recovery at 55 percent and requires a fresh ON request. Manual operation bypasses battery-specific restrictions while retaining other measurement and electrical checks. The backend demonstrates a bounded singly linked list, recursive merge sort and inheritance for interchangeable SQL storage implementations. Live captures show measurements for both load branches and Manual operation at a low battery-feed voltage; automated tests provide additional reproducible evidence. Battery percentage remains approximate, and sensor accuracy, energy savings and electrical compliance have not been established through controlled measurements. The prototype demonstrates an integrated monitoring and control workflow rather than a certified battery-management system.

Keywords: ESP32, solar microgrid, INA219, load prioritization, linked list, recursive merge sort.

Report contents

Section	Page
Problem statement and objectives	3
Requirements and system architecture	4 to 5
Circuit and component selection	6 to 7
Firmware cloud and DSA implementation	8 to 10
Photographs testing and results	11 to 13
Evaluation sustainability and standards	14 to 16
Conclusions costs and team contributions	17 to 18
References and supporting appendices	19 onward

1 Problem statement and measurable objectives

A small off-grid solar installation must distribute limited stored energy among appliances with different priorities. A basic charge controller manages charging, but it does not by itself provide the branch-level website measurements, authenticated remote switching and historical activity record needed for this project. A user should be able to identify which load is drawing current and understand why a requested load is blocked.

We developed a two-load demonstrator in which lighting has higher priority than ventilation. The system measures each branch separately, estimates battery reserve from voltage and applies a local policy before accepting ON requests. The intended users are students and operators studying small DC energy systems; the prototype is not a substitute for a commercial battery-management or protection system.

ID	Objective	Observable acceptance criterion
O1	Measure individual loads	Recognize INA219 addresses 0x40 and 0x41 and report separate valid V, A and W fields.
O2	Provide independent remote control	Each website button operates only its named physical load and receives a device acknowledgement.
O3	Prioritize battery reserve	In Automatic mode, estimated SOC below 50% locks the fan; at least 55% clears its SOC lock without restarting it.
O4	Provide selectable battery automation	Manual bypasses battery rules only; Automatic restores them; reboot starts Automatic with both outputs OFF.
O5	Retain history	Archive valid and invalid samples without inventing readings; provide 24-hour, 7-day and 30-day views and CSV export.
O6	Demonstrate DSA and OOP	Exercise linked-list eviction, recursive sorting and shared SQLite/PostgreSQL archive functionality in tests.

The configured one-second sampling and approximately five-second network and dashboard polling intervals are design settings, not measured latency guarantees. Final channel accuracy, command-delay statistics and endurance targets require a controlled measurement campaign.

2 System requirements

2 1 Hardware and interfaces

Element	Requirement or installed specification
Generation and charging	Declared design rating 20 W; mounted panel rating is not legible in the final photos. Generic 10 A LCD charge controller; no digital controller telemetry.
Energy storage	iVOOMi IV175-7212V VRLA battery, nominal 12 V and 7.2 Ah at the marked C20 rate.
Loads	Listed 12 V 9 W LED bulb and COOLING CLD4010SM 12 V DC fan; actual current and full-charge voltage tolerance need verification.
Controller	Classic ESP32 development board with Wi-Fi; USB power bank supplies its control rail.
Branch sensing	Two INA219 modules, logic supply 3.3 V, bulb address 0x40 and fan address 0x41.
Temperature	DS18B20 powered at 3.3 V, DATA on GPIO4 with a 4.7 kohm pullup.
Actuation	Two active-low 5 V relay channels. Coil/electronics power and switched battery contacts are separate circuits.

2 2 Software and operating conditions

The firmware uses the Arduino ESP32 toolchain, Adafruit INA219 and BusIO, OneWire, DallasTemperature and ArduinoJson. The web service uses Python, FastAPI, Pydantic, SQLAlchemy and a PostgreSQL driver. The browser uses HTML, CSS and JavaScript. A local SQLite mode and an isolated simulation support development without operating real loads.

Cloud operation requires Wi-Fi with internet access, a configured Vercel service, a Supabase database and an authorized user account. Device and operator credentials are separate. The firmware uses certificate-validated HTTPS; device secrets and database credentials are not placed in the public frontend or repository.

The 12 V battery is never a logic-supply input. The assembly uses jumper connections and has no installed fuse holder or fuse in the battery feed. The final cardboard display enclosure is not a certified protective housing. Appropriate DC fault protection, rated conductors and secure terminations remain necessary. Software current limits are not short-circuit protection.

3 System architecture and block diagram

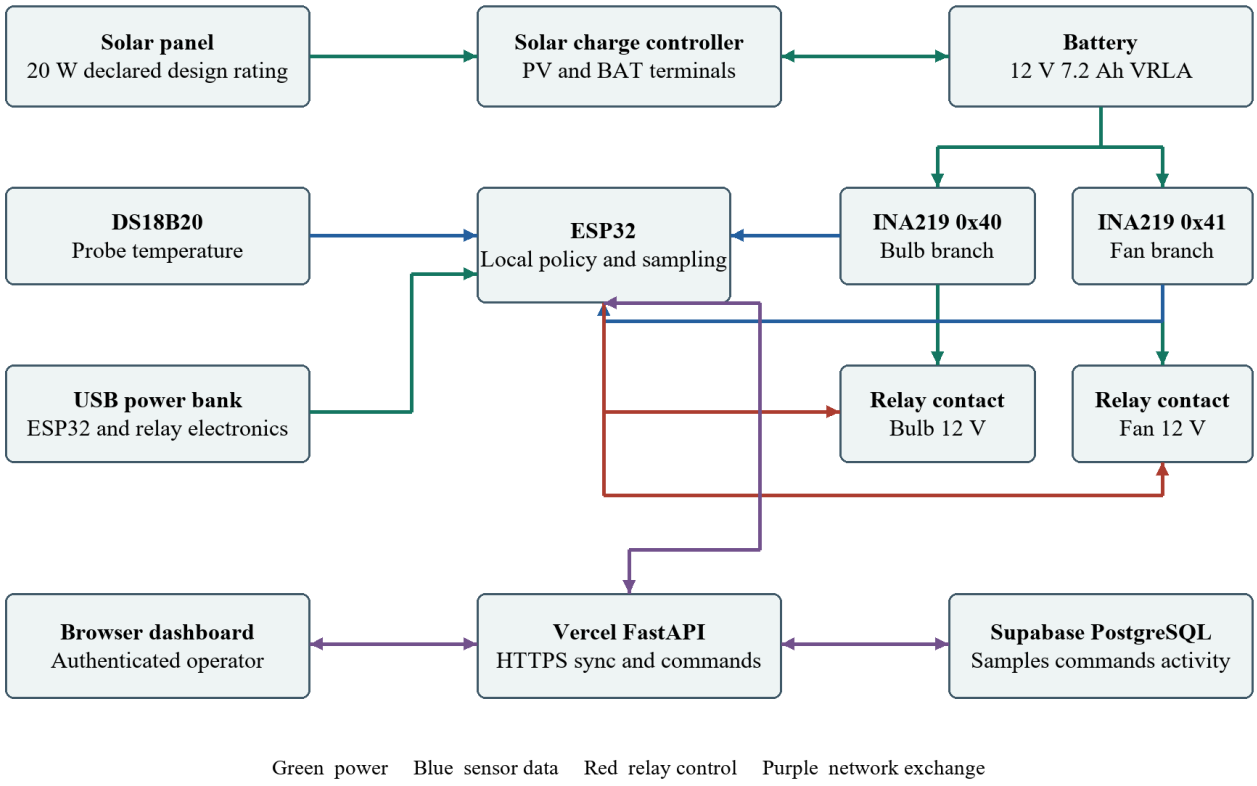


Figure 1 Power sensing control and cloud architecture of the current version

The upper power path connects the panel, charge controller and battery. Two downstream branches feed the bulb and fan through separate current shunts and relay contacts. Each INA219 reports its branch measurements to the ESP32; the temperature probe provides a separate local reading. The power bank supplies the ESP32 and relay electronics, not the 12 V loads.

The ESP32 applies local permission checks and exchanges telemetry, command requests and acknowledgements with the HTTPS backend. Supabase retains durable records, while the browser polls the backend for live and historical views. A closed browser does not stop local firmware policy execution. Internet loss interrupts remote updates and control; it does not itself imply that all loads are switched off.

This architecture replaces the earlier Blynk and serial-collector version. The current two sensors measure load consumption only. Solar generation, net battery charging current and conversion efficiency are not directly measured.

4 Circuit and prototype design

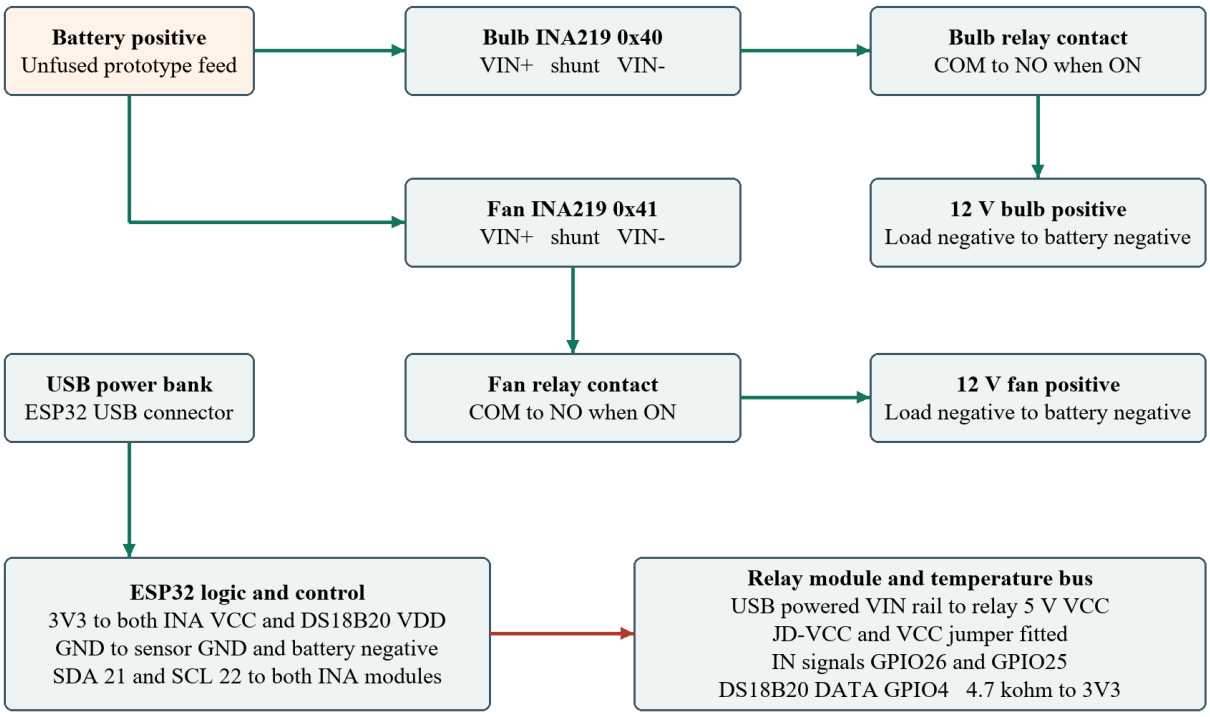


Figure 2 Documented branch topology and logic connections

Connection	Firmware or wiring assignment
Shared I2C bus	SDA GPIO21; SCL GPIO22; both INA219 logic supplies 3.3 V.
Load identities	Bulb sensor 0x40; fan sensor 0x41. VIN- is the outgoing positive shunt terminal, not ground.
Relay outputs	Bulb GPIO26; fan GPIO25; active LOW. Physical contact-to-load identity must match these logical names.
Control power	Power bank to ESP32 USB; USB-powered VIN rail to relay VCC; JD-VCC/VCC jumper fitted; common reference ground.

Sensors upstream of their relay contacts can retain battery-feed voltage while a load is OFF. During development, relay-channel reversal caused a Fan command to operate the bulb while its sensor card correctly followed the measured branch. The software swap was reverted. Final screenshots show both named branches with current and power, but a four-state physical identity acceptance record is still needed; telemetry alone does not establish contact-to-load identity.

Figure 2 records the prototype topology rather than certifying it as safe. The battery feed lacks physical overcurrent protection. No solar PV negative or charge-controller LOAD negative connection is inferred from this diagram. The controller LOAD output is not used for the direct battery-fed load branches.

5 Sensor and actuator selection

5 1 INA219 branch monitors

The INA219 combines shunt-voltage and bus-voltage measurement with an I2C interface. Its 0 to 26 V bus range is suitable for observing a nominal 12 V branch, subject to the breakout module and shunt ratings. Two addressable devices can share the same bus without assigning separate analogue inputs. The selected addresses preserve an explicit correspondence between sensor data and each physical load. [1]

The library calibration used here is the 32 V/2 A configuration for the assumed R100 shunt, with calibration register 4096 and a 100 microampere current increment. The 32 V configuration name does not increase the chip maximum above 26 V. A module with a different shunt requires different calibration. Firmware also checks the actual register-read result, conversion-ready flag, math-overflow flag and calibration value rather than equating an I2C acknowledgement with a valid measurement.

5 2 DS18B20 temperature sensor

The DS18B20 provides digital temperature readings using a 1-Wire interface, reducing analogue conditioning requirements. We use externally powered operation and a pullup resistor on the data line. The manufacturer specifies nominal accuracy of plus or minus 0.5 degrees Celsius over -10 to +85 degrees Celsius; this is a component specification, not a measured accuracy result for our probe. [2]

The probe records temperature at its actual mounting location. It does not establish internal battery temperature or detect every component hotspot. Firmware rejects disconnected readings and the 85 degree initialization sentinel. Its provisional 50 degree trip and 45 degree recovery settings demonstrate thermal hysteresis; a certified thermal-protection design would require validated placement and testing.

5 3 Relay actuator and controller selection

The relay module provides two separately commanded contact paths for the DC loads. Its coil supply is 5 V, while the GPIO signal interface must be checked for reliable 3.3 V operation. Contact ratings, DC switching behavior, fan starting current and wiring capacity must be considered separately from any printed AC relay rating. The ESP32 was selected for integrated Wi-Fi and sufficient digital interfaces.

A USB power bank makes the control section portable without relying on the laptop. This choice does not make the complete demonstrator solar self-powered. A protected, regulated battery-to-control supply is a future integration task, not an installed feature.

6 Firmware and battery control implementation

The firmware samples sensors nominally once per second. A separate network task synchronizes approximately every five seconds, keeping network requests outside the main sensor-and-relay loop. Both relay outputs are initialized OFF. The device reports validity flags, branch measurements, estimated reserve, requested-output state, block reasons and command acknowledgements. [3, 15]

Policy	Configured behavior
Fan reserve rule	Automatic: lock below 50% estimated SOC; release SOC lock at 55%; never restart automatically.
Critical voltage	Automatic: both loads blocked at or below 11.8 V; critical latch recovery at or above 12.1 V.
Temperature	Both modes: provisional trip at 50 degrees C; recovery at 45 degrees C.
Branch current	Both modes: provisional bulb 1.5 A and fan 0.3 A limits; trip latch requires an OFF reset when conditions permit.
Supply validity	Both modes: missing measurements block ON; provisional supply maximum 12.8 V; polarity and register errors remain faults.
Mode and restart	Manual bypasses low-voltage and fan-SOC rules only. Reboot restores Automatic and both outputs OFF.

Battery voltage is selected only from valid branch-source readings between 8 and 16.5 V; two qualifying readings differing by more than 0.25 V are rejected. A successful zero-volt sensor read is a valid branch measurement but cannot supply a battery estimate. This avoids confusing an unpowered branch with failed communication.

For display and the fan rule, voltage is filtered using 20 percent of the new reading and 80 percent of the previous filtered value. A generic lookup maps 11.8, 12.0, 12.3, 12.6 and 12.8 V to 0, 25, 50, 75 and 100 percent, with interpolation. Raw source voltage is retained for immediate voltage checks. The lookup is not calibrated to actual battery capacity.

Mode revisions and command identifiers prevent an older ON request from being applied after a mode change. The protocol reports applied or rejected commands explicitly. It does not include a mechanical contact-position sensor, so an acknowledgement is confirmation of firmware action rather than independent proof of relay closure.

7 Website backend and historical storage

The browser provides an overview, independent load controls, an Automatic protection switch, historical graphs and an activity view. Its live-data request runs approximately every five seconds and history refreshes approximately every minute. These polling intervals provide near-real-time updates; they are not a continuous streaming channel or a measured end-to-end response-time guarantee. [15]

FastAPI accepts authenticated dashboard requests and independently authenticated device synchronization. The backend validates structured telemetry and stores samples, command records, boot identifiers and activity in the private urjagrid PostgreSQL schema. Supabase provides durable storage across Vercel serverless instances. Local development uses a distinct SQLite archive. [4, 5, 6]

Record or operation	Purpose
ug_samples	Time-stamped per-load measurements, validity, mode and interval-energy values.
ug_devices and ug_boots	Latest device state, last-seen time and boot/sequence tracking.
ug_commands and ug_command_context	Expiring requests, actors, status, acknowledgements and expected mode revision.
ug_activity	Command events, faults and maintenance notes for the activity timeline.
History and export	24-hour, seven-day and rolling 30-day views; CSV and browser print/PDF output.

Duplicate or out-of-order samples do not replace the newest device state. ON requests are not left waiting indefinitely for an offline device: commands expire after 15 seconds, and device online status uses a 20-second freshness threshold. A reboot invalidates requests tied to the preceding boot. The local database is not automatically synchronized with the cloud database.

Power is computed from voltage and current. Historical watt-hours are estimated by trapezoidal integration between valid adjacent samples; intervals exceeding 30 seconds and transitions across boots are not treated as continuously measured consumption. History graphs aggregate into at most 720 time buckets and display coverage, preventing a sparse record from being presented as complete energy metering.

Hosted user access uses Supabase authentication and an email allowlist. Separate device credentials, exact-origin checks and HTTPS protect the control interface. Secrets remain in ignored local configuration or hosting settings. No security or penetration-test certification is claimed. [3, 5]

8 Data structures algorithms and inheritance

8 1 Singly linked list

EventNode stores an event dictionary and a reference named next. RecentEventList maintains head, tail, size and a capacity of 200. Appending a node updates tail.next and then the tail reference. If capacity is exceeded, pop_oldest advances head to head.next; the newest 200 records remain. snapshot traverses the nodes and returns dictionaries suitable for JSON serialization. [15]

In the current cloud backend, the linked list is reconstructed from committed database activity records while producing a history response. It is an in-memory presentation structure, not permanent storage or a cross-instance serverless cache. This is different from the retained legacy serial backend, which maintained a process-local event cache.

Operation	Time complexity	Reason
Append event	O(1)	Tail already identifies the insertion point.
Remove oldest event	O(1)	Only the head reference changes.
Snapshot traversal	O(n)	Each retained node is visited once.
Recursive merge sort	O(n log n)	Each level merges n items over logarithmically many levels.

8 2 Recursive merge sort

recursive_merge_sort returns immediately for zero or one record, recursively sorts the left and right halves and merges them using two indices. Choosing the left item on equal keys preserves stable ordering. Activity records and notes are sorted by timestamp and record identifier in descending order. The function does not call Python sorted or list.sort; tests compare its output against a reference sort, including randomized inputs and ties.

8 3 Classes inheritance and storage reuse

ArchiveBase defines the storage interface. SQLArchive implements shared SQL-backed synchronization, history and command handling. SQLiteArchive and PostgresArchive inherit that implementation while supplying their database-specific connection setup. Inheritance therefore shares functionality and reduces duplicated code; it does not act as a data-transfer pipeline between databases.

The linked list uses O(n) node storage. Merge sort has O(n) peak auxiliary storage and O(log n) recursion depth for fixed-size event records. These choices provide explicit educational examples; this project does not claim that linked lists outperform a standard deque, or that the relay policy is implemented as a heap or priority queue.

9 Prototype photographs and dashboard evidence

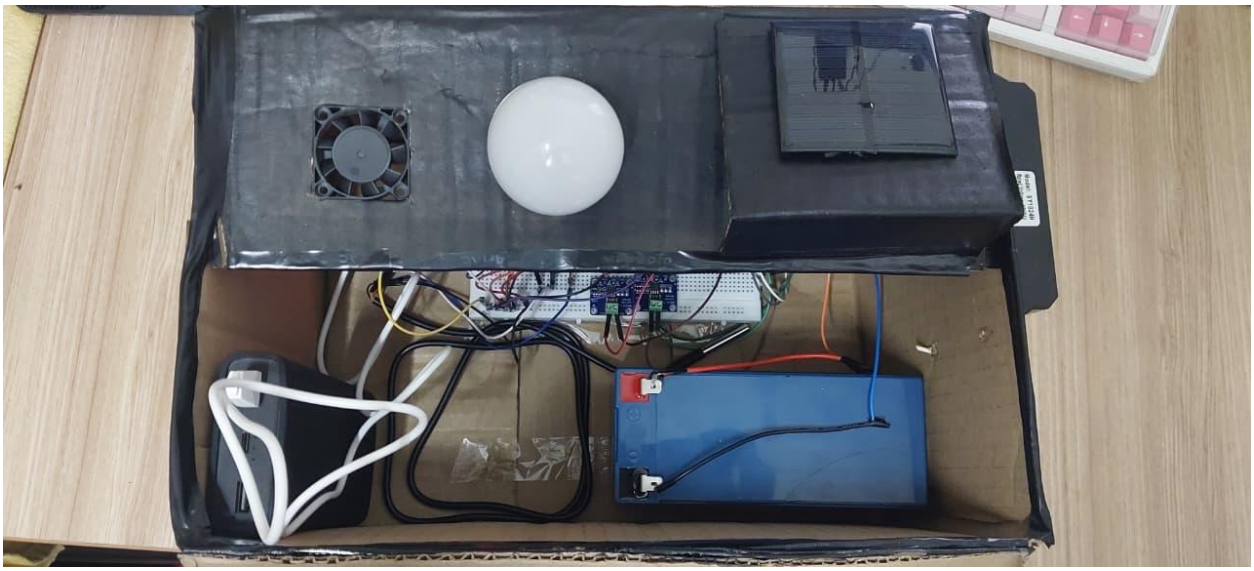


Figure 3 Final enclosure open to show the battery breadboard sensors and USB power bank

The removable lid carries the fan, bulb and mounted solar panel. The base contains the battery and control wiring, with the USB power bank alongside. The arrangement demonstrates a compact exhibition assembly; exposed wiring and cardboard construction do not establish electrical or fire safety.



Figure 4 Front mounted charge controller with an LCD reading of approximately 10.6 V

These photographs were supplied on 27 October 2026. The overall external view appears in Appendix C. The controller reading and website readings were not obtained as a synchronized calibration pair. Appendices D and E reproduce the live overview, individual load cards, history graphs and command log.

10 Test procedure and calibration

10 1 Staged functional procedure

Initial checks separate low-voltage logic from the battery and solar panel. With USB-only power, inspect sensor supplies and common reference, scan for addresses 0x40 and 0x41, confirm temperature communication and check that both relay outputs start OFF. An address acknowledgement alone is insufficient: firmware must also report successful register reads and valid conversion status.

Before a powered load test, verify polarity, secure terminations, load ratings and an appropriately protected supply. Energize one branch at a time and record which physical load operates. For each ON and OFF request, record the named dashboard card, observed load, branch V/A/W, acknowledgement and time. Repeat the mapping check after any IN1/IN2 wiring change. No destructive short-circuit test is part of this procedure.

Exercise reserve transitions with the isolated software simulation rather than deliberately deep-discharging the real battery. Test 45%, 52% and 60%, then request a fresh fan ON. Test Manual at low estimated reserve, restore Automatic and check the resulting OFF action. A hardware policy test requires a controlled source or safe replay arrangement and independent observation; a simulated pass does not substitute for it.

10 2 Calibration and measurement uncertainty

The implemented INA219 calibration assumes the module shunt value and configures the conversion scale; this is not a traceable instrument calibration. For metrology, compare each branch with a reference voltmeter across the same measurement points and a suitably rated series ammeter at several stable loads. Record zero-current offset, reference readings, indicated values, ambient conditions and repeated samples. No gain correction should be fitted to unsynchronized measurements taken at different points.

Compare the temperature probe against a reference thermometer at stabilized temperatures and the same location. A precision current source, calibrated thermometer, shunt tolerance record and instrument uncertainty certificate were not part of the documented demonstration. Therefore a numerical sensor-accuracy or battery-SOC accuracy result is not established.

Quantity	Required calibration record	Current status
Branch voltage and current	Paired reference/INA values at identical operating points	To be recorded
Probe temperature	Paired stabilized thermometer/probe values	To be recorded
True battery reserve	Controlled capacity or charge-accounting reference	Not established
Software correctness	Repeatable unit and host-policy tests	Passed on 27 October 2026

11 Recorded results and software verification

The live screenshots dated 27 October 2026 provide the following displayed measurements. Both sensor cards show available readings and a device-reported ON state. These are dashboard observations, not reference-instrument calibration results. The overview and load-card captures are separate frames and must not be treated as exactly simultaneous.

Quantity	Bulb 0x40	Fan 0x41
Branch supply voltage	9.51 V	9.54 V
Current	0.221 A	0.075 A
Displayed power	2.10 W	0.72 W
Device-reported state	ON	ON
Most recent request status in capture	Applied	Expired

The calculated products are 9.51 x 0.221 = 2.10171 W and 9.54 x 0.075 = 0.7155 W, consistent with the rounded branch displays. Their rounded sum is 2.82 W; the separate overview shows 2.81 W. This small difference is consistent with different frame times or rounding and is not an accuracy assessment.

Overview or history field	Displayed result
Battery feed and estimated reserve	9.55 V and 0% estimated SOC
Temperature and total load	25.2 degrees C and 2.81 W
Confirmed operating mode	Automatic OFF with Manual battery override active
Selected history window	71 samples; bulb energy 0.03 Wh; fan 0.00 Wh; combined 0.03 Wh
Measured-time coverage	Bulb 0.5%; fan 0.6% of the selected 24-hour window

The 9.55 V observation is below the configured 11.8 V critical threshold. Load operation in this capture reflects Manual bypass, not successful Automatic battery protection. Continued discharge under these conditions is not a recommended operating mode. The fan request expiry remains a command-delivery limitation even though the latest device state reports ON.

Earlier troubleshooting produced multimeter readings of 11.69 V under connected conditions and 11.88 V after load removal, plus a stored temperature of 24.75 degrees C. Different times and measurement points prevent direct calibration comparisons. All 42 Python tests and both C++ host suites passed on 27 October 2026; these tests did not operate real relays.

12 Evaluation against objectives and limitations

Objective	Available verification	Assessment
O1 separate sensing	Final load cards show valid separate V/A/W for 0x40 and 0x41.	Demonstrated; reference calibration and repeated trials pending.
O2 independent switching	Both loads report ON; bulb request applied. Other requests expired or were rejected.	Partly demonstrated; mapping acceptance and delivery reliability require checks.
O3 reserve prioritization	C++ and simulation checks cover cutoff, recovery and no auto-restart.	Verified in software; physical threshold trace pending.
O4 selectable mode	Tests pass; final overview confirms Automatic OFF and Manual operation.	Manual mode demonstrated; physical Automatic transition trace pending.
O5 cloud and history	71 archived samples shown with graph, energy and low coverage; archive tests pass.	Demonstrated for a short session, not continuous 24-hour coverage.
O6 DSA and inheritance	Linked-list, recursive-ordering and archive/API regression tests pass.	Demonstrated in implementation and tests.

Voltage-derived reserve is the central measurement limitation. Charging voltage, surface charge, temperature, ageing and load-induced voltage drop can change the estimate without a proportional change in stored charge. The two branch sensors do not measure net charging current, so they cannot support complete battery coulomb counting or solar-yield measurement.

Physical limitations include unfused jumper wiring, a cardboard display housing, unverified load tolerance at the battery label charging range of 14.5 to 15.0 V, uncertain probe coupling and no relay-contact feedback. The 12.8 V software ceiling may block operation while charging; raising it without verified load ratings is not a solution. The photographed panel rating is not legible, so the declared 20 W design rating is not independently confirmed.

Cloud polling introduces delay and depends on internet access. The supplied command log includes applied, expired and rejected requests; a rejected entry states that reported state or mode differs from the request. This documents incomplete acknowledgements, not guaranteed command success. A Manual mode persists through Wi-Fi loss until changed or rebooted. No unattended safety, latency, runtime or energy-efficiency certification is claimed.

13 Sustainable Development Goal contribution

13 1 SDG 7 Affordable and Clean Energy

The project relates most directly to renewable-energy use and informed energy consumption under SDG 7. Target 7.2 concerns increasing the renewable share, while target 7.3 concerns improving energy efficiency. Our panel-and-battery demonstrator provides a small-scale setting for studying renewable supply, and its branch measurements expose where stored energy is consumed. [7]

The specific contribution is a working mechanism for reserving energy for a higher-priority lighting load while restricting the fan at low estimated reserve. The history view can support comparisons of operating schedules. However, load shedding reduces service or consumption; it does not by itself prove improved conversion efficiency. We have not measured solar yield, energy savings against a baseline or improved access to electricity, so the contribution is an educational demonstration rather than a quantified SDG impact.

13 2 SDG 9 Industry Innovation and Infrastructure

Target 9.5 emphasizes research and technological capability. Our contribution is the design and integration of an addressable sensor network, a microcontroller policy, authenticated remote commands and persistent data storage. The source repository and reproducible tests make the implementation inspectable and reusable for subsequent student work. [8]

The prototype shows how a small energy system can expose operating data and implement local decisions even when a remote page is closed. It is not deployed public infrastructure, a utility-grade controller or evidence of an economy-wide increase in industrial capability. The technically specific output is a tested educational architecture, not a claim to have achieved the global indicator.

13 3 How future impact would be measured

A follow-up study should compare identical operating periods with and without the reserve policy, documenting load service time, watt-hours, valid-data coverage and battery conditions. Separate generation and battery-current measurements would allow renewable energy and charging losses to be assessed. Battery replacement and disposal practices must also be addressed before claiming environmental benefit over the product life cycle.

SDG 3 and SDG 11 appeared in earlier presentation artwork, but no health outcome or city-level deployment was measured. They are not presented here as directly demonstrated contributions. A future rural-lighting application remains an application prospect, not a completed field installation.

14 Applicable BIS standards and check status

We searched official BIS catalogue material, laboratory test listings and the public solar-battery test-report form using the terms solar photovoltaic battery, PV power converter, electrical wiring and energy management. The following references apply at component, installation or management-system level. This review identifies relevant checks; it is not a complete clause audit or proof of certification. [9 to 13]

14 1 Solar battery and converter requirements

IS 16270:2023, Secondary cells and batteries for solar photovoltaic application - General requirements and methods of test, is relevant to the VRLA battery used for solar storage. The BIS test form identifies deep-discharge protection (5.3), marking (5.4), safety (5.5), documentation (5.6), capacity (8.1), endurance (8.2) and charge retention (8.3). We checked photographed nameplate information and implemented reserve-related logic; we did not perform those standardized battery tests. The 7.2 Ah capacity remains a label rating. [9, 10]

IS 16221 (Part 1):2016, Safety of Power Converters for Use in Photovoltaic Power Systems Part 1 General Requirements, is relevant to assessing the purchased solar charge controller as PV power-conversion equipment. BIS listings cover marking, environmental conditions, electrical and fire hazards, components and safety-related firmware. No controller conformity report, insulation test, thermal test or abnormal-condition test was obtained. Relay software tests do not establish converter compliance. Part 2 inverter requirements are not asserted because no AC inverter is installed. [11]

14 2 Wiring and energy management references

IS 732:2019, Code of Practice for Electrical Wiring Installations, provides an installation-design and verification reference for selecting conductors, connections, isolation and protective arrangements. We reviewed the functional topology and separated logic supplies from battery power. The present unfused jumper-wire assembly has not passed a standards-based wiring inspection, fault-protection test or installation verification. [12]

IS/ISO 50001:2018, Energy management systems - Requirements with guidance for use, is relevant as an organizational energy-management framework. Logging branch consumption and reviewing history can support monitoring and energy review. No formal energy baseline, organizational management-system audit or certified improvement programme was completed. It is not a product-safety certificate. [13, 14]

Status summary: nameplate and functional checks are partial observations; standardized capacity, electrical-safety and management-system conformity checks are not completed. The earlier poster IEC references are not silently relabeled as BIS certification. No BIS mark or certificate is claimed for the assembled demonstrator.

15 Conclusions and future development

UrjaGrid Monitor integrates two separately addressed load sensors, a temperature probe, an ESP32 and independent relays with a hosted control interface and persistent archive. The implementation makes load permission explicit through block reasons and device acknowledgements. It also connects data-structure exercises to actual report preparation rather than placing unrelated example algorithms beside the application.

Live captures now demonstrate separate measured branches: the bulb card displays 9.51 V, 0.221 A and 2.10 W, while the fan displays 9.54 V, 0.075 A and 0.72 W. The overview reports Manual mode, a 9.55 V battery feed and 25.2 degrees C. Forty-two Python tests and both C++ suites also passed. These results demonstrate integration, not calibrated accuracy or a valid remaining-capacity measurement.

The reserve policy is intentionally conservative about restarting loads. Automatic mode can disconnect the fan below the configured estimated threshold, but recovery only restores permission. A fresh operator request is required. Manual mode makes battery-policy bypass visible while retaining other checks, and a reboot returns the controller to Automatic with both loads OFF. This separation improves the clarity of the demonstration, although a Manual override still risks excessive discharge.

15 1 Priorities for the next version

The first priority is to stop relying on Manual operation at a deeply depleted battery-feed voltage. Add suitable DC fault protection, rated wiring, secure terminals, a protective housing and verified supply compatibility. The finished cardboard display enclosure improves presentation but is not certified protection. Confirm relay-to-load identity and check full-charge load compatibility before an unattended run.

The next measurement priority is synchronized calibration against reference instruments, followed by a stable recording session for all four load states. A dedicated battery-current or charge-accounting monitor and temperature-aware battery characterization would improve reserve estimation. A generation-side sensor would be needed to measure solar production; neither existing load sensor should be relabeled as a solar sensor.

The supplied history view has only 0.5 to 0.6 percent coverage, and several commands expired or were rejected. Further work should characterize command latency, investigate missing acknowledgements and collect a sustained calibrated recording. Backup and retention policies also need review. Any local-to-cloud database synchronization requires a separately designed migration or replication mechanism.

The project is therefore a reproducible educational foundation for studying DC energy monitoring and load prioritization. Its claims remain limited to the functions and evidence documented in this report; electrical certification and quantified field benefit require further work.

16 Bill of materials and individual contributions

The team reports total expenditure of INR 6,300. Three ESP32 boards were bought at INR 450 each, costing INR 1,350; one is installed. Other line amounts are reference estimates, not an invoice reconstruction. Appendix F explains sources and reconciliation.

Component	Bought / used	Unit / line INR	Basis
ESP32 boards	3 / 1	450 / 1,350	Confirmed
20 W panel comparator	1 / 1	1,731 / 1,731	Estimate [16]
10 A LCD controller	1 / 1	389 / 389	Estimate [16]
12 V 7.2 Ah battery	1 / 1	1,399 / 1,399	MRP proxy [23]
INA219 modules	2 / 2	149 / 298	Estimate [17]
DS18B20 probe	1 / 1	68.44 / 68.44	Estimate [18]
Two-channel 5 V relay	1 / 1	74 / 74	Estimate [19]
12 V bulb	1 / 1	209 / 209	Capture [23]
4010 12 V fan comparator	1 / 1	85 / 85	Estimate [20]
Breadboard and wires	1 set	69 + 89 / 158	Estimate [21,22]
Housing power bank cables extras	1 set	Not itemized	No receipt split

Priced reference lines total INR 5,761.44. The INR 538.56 difference from reported spend is not allocated by receipt. Panel and fan references are comparators; list prices are not historical paid amounts.

16 1 Confirmed individual contributions

Member	Contribution confirmed by the team
Akshat Srivastava	Wiring, data and power flow, project planning.
Prajit Yadav	DSA planning, frontend and backend software.
Uppala Mahidhara Venkata Trivikram	Enclosure design.

Proposed follow-up allocation: Akshat - calibration records; Prajit - command and data analysis; Trivikram - figure captions and report checks. These allocations do not claim additional work already completed.

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[10] Bureau of Indian Standards. Guidelines for implementation of IS 16270:2023 dated 26 March 2025.

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[14] International Organization for Standardization. ISO 50001:2018 Energy management systems requirements with guidance for use.

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[15] Synchron Team 606. UrjaGrid Monitor source code and software verification record.

https://github.com/akshatsri121/UrjaGrid-Monitor

Technical references checked 27 October 2026. Catalogue and test-scope records do not replace a full-standard clause review.

Code and demonstration links

Code: https://github.com/akshatsri121/UrjaGrid-Monitor

Dashboard: https://urja-grid-monitor.vercel.app

Demonstration video unlisted: https://youtube.com/shorts/6TPqgRCmAmI

Component price references

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https://mechatronlab.com/product/ina219-current-and-power-sensor

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https://robu.in/product-category/breadboard-and-zero-pcb/

[22] Robu. Jumper-wire category listing 65-piece flexible wire pack INR 89.

https://robu.in/product-category/dupont-cable/

[23] Synchron Team 606. Battery photograph showing INR 1399 MRP; supplied bulb listing capture showing INR 209; reported total INR 6300 and ESP32 purchase amounts. Private project records, not invoices.

Price references checked 28 October 2026. Listings are indicative and subject to tax presentation, availability and shipping changes. They are not receipts for the reported INR 6300 total.

Appendix A Selected implementation code

The excerpts below identify the implemented DSA structures. Complete firmware, backend and tests are linked in reference [15]. Configuration secrets are intentionally excluded.

A 1 Linked list insertion

```
def append(self, event):
    node = EventNode(dict(event))
    if self.tail is None:
        self.head = node
    else:
        self.tail.next = node
    self.tail = node
    self.size += 1
    if self.size > self.capacity:
        self.pop_oldest()
```

A 2 Recursive division and merge

```
def recursive_merge_sort(items, key, reverse=False):
    if len(items) <= 1:
        return list(items)
    midpoint = len(items) // 2
    left = recursive_merge_sort(items[:midpoint], key, reverse)
    right = recursive_merge_sort(items[midpoint:], key, reverse)
    merged = []
    i = j = 0
    while i < len(left) and j < len(right):
        take_left = (key(left[i]) >= key(right[j]) if reverse
                     else key(left[i]) <= key(right[j]))
        if take_left:
            merged.append(left[i]); i += 1
        else:
            merged.append(right[j]); j += 1
    merged.extend(left[i:])
    merged.extend(right[j:])
    return merged
```

Source: backend/data_structures.py. The conditional expression is line-wrapped here for readability; the sorting and insertion behavior is unchanged.

Appendix B Sample output and completion records

B 1 Software test output

Re-run on 27 October 2026 using temporary databases and host executables. No COM port or physical relay command was used.

Python unittest discovery
Ran 42 tests
OK

Firmware policy tests passed
Automatic and Manual modes; immediate protection restoration
SOC hysteresis; no automatic restart; retained interlocks

INA219 tests passed
Verified zero readings; bus faults; calibration; conversion flags
Ranges; battery-source selection; retained interlocks
Commands: python -m unittest discover -s backend -q; compile and run firmware/tests/control_policy_test.cpp and firmware/tests/sensor_validation_test.cpp with a C++11 compiler. The Python run emitted a Starlette/httpx deprecation warning; all 42 tests completed successfully.

B 2 Final measurement record to complete

State	Bulb V A W	Fan V A W	Reference and time
Both OFF	Not recorded	Not recorded	No reference trial
Bulb only ON	Not recorded	Not recorded	No reference trial
Fan only ON	Not recorded	Not recorded	No reference trial
Both reported ON	9.51 / .221 / 2.10	9.54 / .075 / .72	27 Oct dashboard; Manual

The last row transcribes the supplied screenshot, not an endorsed low-voltage test. Further controlled trials require a healthy protected supply. Record each state with a reference instrument and repeated samples before deriving accuracy or latency statistics.

B 3 Submission attachments

Final photographs, live screenshots, reported expenditure, confirmed roles and the supplied video link are included. Reference calibration and an institution-approved plagiarism or similarity report remain separate submission tasks. No similarity percentage or certificate is claimed.

Appendix C External enclosure and assembly



Figure 5 External front view with lifted lid and mounted controller

The display housing accommodates the bulb, fan and mounted solar panel on its upper surface and places the controller on the front. The liftable section exposes the internal wiring for demonstration. Trivikram designed the enclosure. Photographs document its arrangement; they do not establish flame resistance, ingress protection or cable strain relief.

The controller reads approximately 10.6 V in these photographs. The later dashboard frames show a lower branch supply under different conditions. Without matched timestamps, common test points and reference readings, this difference is not evidence of measured sensor error.

Appendix D Live overview and independent load cards

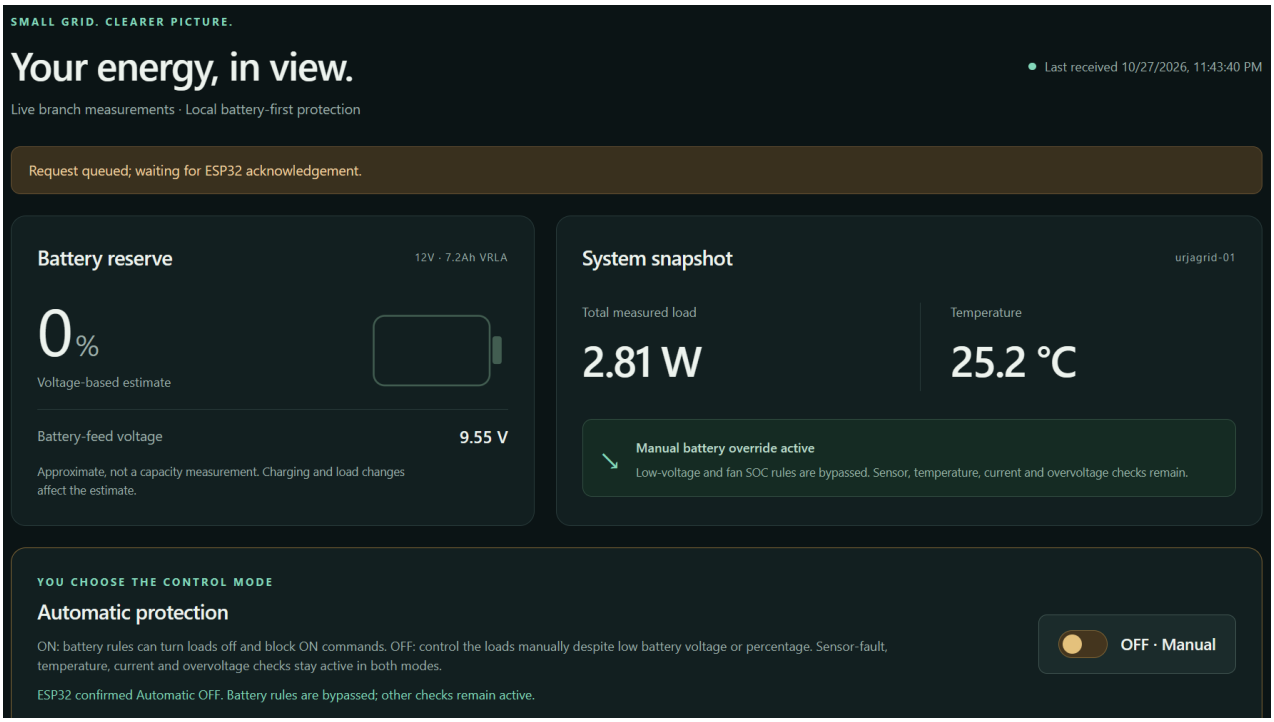


Figure 6 Live overview with 9.55 V battery feed and confirmed Manual battery override

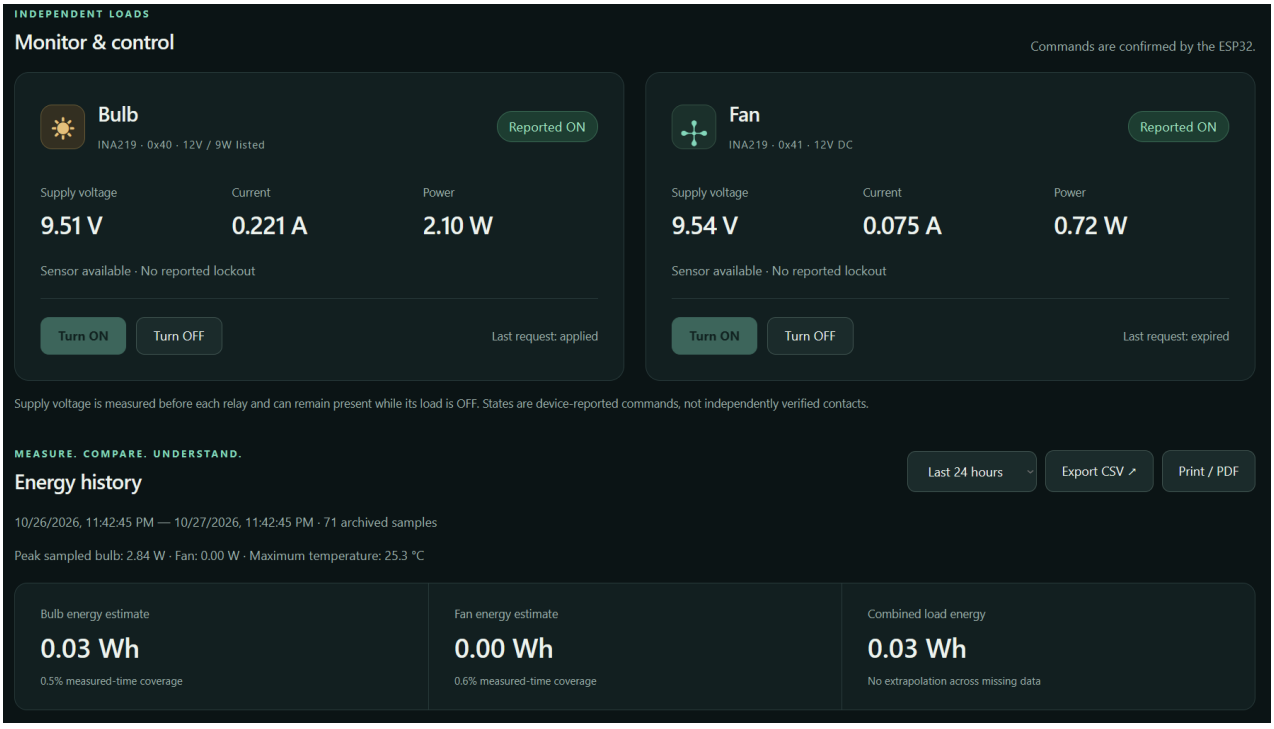


Figure 7 Separate load readings with an applied bulb request and expired fan request

Screenshots supplied on 27 October 2026. Both channels report ON and available sensor readings. Manual mode explains operation below the Automatic critical-voltage threshold; it does not establish healthy battery reserve or recommended continuous operation.

Appendix E Historical graphs and command audit

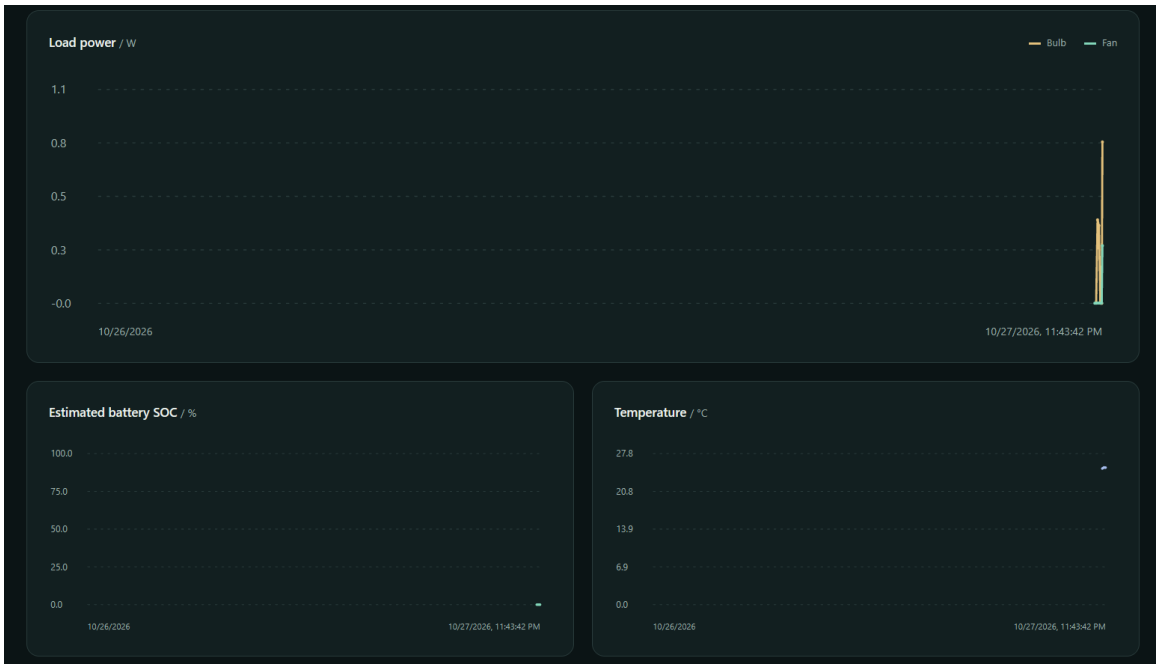


Figure 8 Short recorded session within the selected 24 hour history window

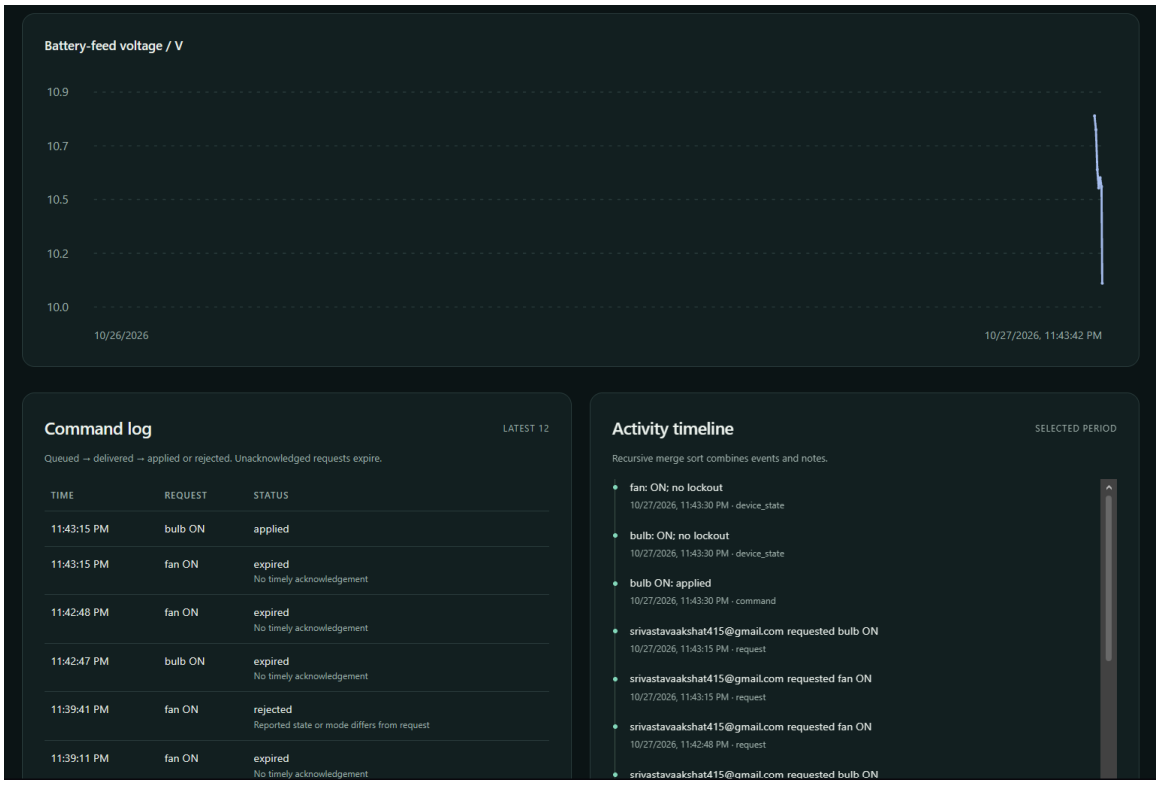


Figure 9 Battery feed trend and command log with applied expired and rejected requests

Most plotted activity is concentrated near the right edge. This is not a full day of continuous valid data. The history summary shows only 0.5 to 0.6 percent measured-time coverage. Expired requests indicate that timely matching acknowledgement was not recorded; device-reported state and request outcome are separate observations.

Appendix F Cost basis and expenditure reconciliation

The reported project spend is INR 6300, including three ESP32 purchases at INR 450 each. Only one board is installed. Thus INR 900 of confirmed board expenditure is outside the installed single-board configuration; the condition of the two other boards is not inferred.

Cost category	Amount INR	Evidence basis
Confirmed overall spend	6,300.00	Team statement
Confirmed ESP32 purchases	1,350.00	3 x 450; one installed
Remainder of actual spend	4,950.00	6300 minus 1350; not split by receipt
Priced reference-line total	5,761.44	Confirmed ESP32 line plus estimated or proxy lines
Unreconciled reference difference	538.56	6300 minus 5761.44; not an attributed actual charge

The INR 1731 panel reference is for a comparable listed 20 W product, not proof of the rating or price of the small panel visible on the enclosure. The INR 85 fan reference is a 4010 12 V comparator, not an exact CLD4010SM invoice. The battery INR 1399 figure is a photographed MRP proxy, not a paid price. The bulb INR 209 figure comes from the earlier supplied product capture; its shortened Amazon link could not be fetched during revision.

The controller estimate uses the Robu category listing for the product family linked by the team. INA219 and DS18B20 estimates use current module listings. Relay and fan prices were read from public vendor product data; the fan was unavailable. Breadboard and wire amounts use generic comparable items. Shipping, housing materials, the power bank, USB cable, resistor and other extras have no confirmed individual allocation.

The benchmark subtotal and reported actual expenditure are presented separately because receipt-level reconciliation is not available. Installed-build replacement cost and historical project spend are different quantities. The project total includes all three ESP32 purchases; the circuit description includes only the installed controller.

The unused buck converter remains outside the installed design. Its purchase cost, any protective parts not installed and any cloud-service charges have not been assigned individual amounts without supporting records.

Team controller link: https://robu.in/product/10a-intelligent-lcd-solar-controller/

Team bulb link: https://amzn.in/d/0fSiWpCJ

UrjaGridMonitor_Project_Report

by Akshat Srivastava

General metrics

46,997	6,506	784	26 min 1 sec	50 min 2 sec
characters	words	sentences	reading time	speaking time

Score

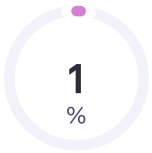


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7	Ungrammatical sentence	
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5	Improper formatting	
1	Conjunction use	
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